Bell Flight Internship Report

Prepared by: Ashwin Shankar

October 15, 2026

Abstract

This report summarizes the week-wise study of aircraft fundamentals, fuel systems, fire protection systems, rotorcraft avionics, navigation systems, communication systems and aircraft electrical power systems. The report explains the important concepts studied each week and presents relevant comparisons, advantages, disadvantages, equations and block diagrams.Simulations were performed using PLECS.

Contents

1 Week 1: Aircraft Aerodynamics, Fuels, Fuel Tanks, Aircraft Parts and Fuel Systems 4
1.1 Aerodynamics of an Aircraft . . . 4
1.1.1 Lift . . . 4
1.1.2 Weight . . . 4
1.1.3 Thrust . . . 4
1.1.4 Drag . . . 4
1.1.5 Condition for Steady Level Flight . . . 4
1.2 Aviation Fuels . . . 5
1.2.1 Jet A . . . 5
1.2.2 Jet A-1 . . . 5
1.2.3 Avgas . . . 5
1.2.4 Biofuel and Sustainable Aviation Fuel . . . 5
1.2.5 Fuel Selection Criteria . . . 5
1.3 Types of Fuel Tanks in Aircraft . . . 5
1.4 Parts of an Aircraft . . . 6
1.4.1 Fuselage . . . 6
1.4.2 Nose . . . 6
1.4.3 Empennage . . . 6
1.4.4 Horizontal Stabiliser . . . 6
1.4.5 Vertical Stabiliser . . . 6
1.4.6 Elevator . . . 6
1.4.7 Rudder . . . 6
1.4.8 Ailerons . . . 6
1.4.9 Flaps . . . 6
1.4.10 Slats . . . 6
1.4.11 Nacelles . . . 6
1.4.12 Stabilator . . . 7
1.4.13 V-Tail . . . 7
1.4.14 T-Tail . . . 7
1.5 Aircraft Fuel System . . . 7
1.5.1 Fuel Pickup / Outlet . . . 7
1.5.2 Sump and Drain . . . 7
1.5.3 Vent Lines . . . 7
1.5.4 One-Way Valves . . . 7
1.5.5 Cross-Feed Manifold . . . 7
1.5.6 Flexible Hose Assemblies . . . 7
1.5.7 Rigid Aluminium and Stainless Steel Tubing . . . 7
1.6 Filters Used in Aircraft Fuel Systems . . . 8
1.7 Fuel Pumps . . . 8

2 Week 2: Aircraft Fire Protection Systems **8**
2.1 Introduction . . . 8
2.2 Smoke Sensors . . . 8
2.3 Flame Sensors . . . 9
2.3.1 Block Diagram of a Flame Sensor . . . 9
2.4 Heat Sensors . . . 9
2.5 Handheld Fire Extinguishers . . . 9
2.6 Aircraft Fire Zones . . . 9
2.7 Types of Extinguishing Agents . . . 10
2.8 Regulatory Standards for Testing and Maintenance . . . 10
2.9 Safety Standards . . . 10

3 Week 3: Rotorcraft Avionics, Navigation and Communication Systems **10**
3.1 Unique Requirements of Rotorcraft Avionics . . . 10
3.2 TAWS: Terrain Awareness and Warning System . . . 10
3.2.1 TAWS Architecture . . . 11
3.3 HUMS: Health and Usage Monitoring System . . . 11
3.3.1 HUMS Architecture . . . 11
3.4 VHF, VOR, VOR/DME, VOR Navigation and VORTAC . . . 11
3.5 JPALS . . . 11
3.6 ARA, PinS, RNAV/RNP and Doppler-Based Navigation . . . 11
3.6.1 Doppler Shift Equation . . . 12
3.7 Communication Systems in Rotorcraft . . . 12
3.8 Automatic Direction Finder . . . 13

4 Week 4: Aircraft Electrical Systems **13**
4.1 Introduction . . . 13
4.2 DC and AC Power Systems . . . 13
4.3 Power Conversion . . . 13
4.3.1 Rectifier . . . 13
4.3.2 Inverter . . . 13
4.3.3 Transformer . . . 13
4.4 Generator and Alternator . . . 14
4.4.1 Generator . . . 14
4.4.2 Alternator . . . 14
4.5 Power Distribution Buses . . . 14
4.6 Priority Order During Engine or Alternator Failure . . . 15
4.7 Electrical Load Management System . . . 15
4.8 Ground Power Unit Architecture . . . 15
4.8.1 GPU Block Diagram . . . 16
4.9 External Power Receptacle . . . 16
4.10 Essential and Non-Essential AC and DC Consumers . . . 16
4.11 Pros and Cons of Aircraft Electrical System Elements . . . 16

5 Week 5:Design and Simulation of a Ground Power Unit **17**
5.1 Objective . . . 17
5.2 System Architecture . . . 17
5.3 Schematic Diagram . . . 17
5.4 Design Specifications . . . 17
5.5 Rectifier and DC-Link Stage . . . 18
5.6 Three-Phase Inverter Stage . . . 18
5.7 AC Output Filter . . . 18
5.8 Buck Converter for 28 V DC Output . . . 19
5.9 Total Harmonic Distortion . . . 19
5.10 Simulation Parameters . . . 19
5.10.1 AC Output Results . . . 20
5.10.2 28 V DC Output Results . . . 20

5.11 Simulation Waveforms 21
5.12 Expected and Actual Results 26
5.13 Discussion 27
5.14 Conclusion 27

6 Week 6:Simulation of Power Conversion systems in Aircrafts including APU 27

7 Conclusion 27

1. Week 1: Aircraft Aerodynamics, Fuels, Fuel Tanks, Aircraft Parts and Fuel Systems

1.1. Aerodynamics of an Aircraft

Aerodynamics is the study of how air flows around an aircraft and how this airflow produces forces on the aircraft. In steady level flight, the aircraft is mainly affected by four forces: lift, weight, thrust and drag.

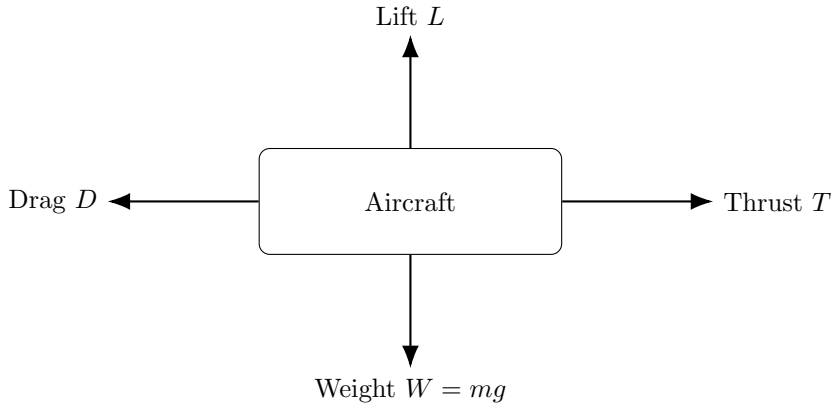


Figure 1: Four fundamental forces acting on an aircraft

1.1.1. Lift

Lift is the upward aerodynamic force produced mainly by the wings. It acts approximately perpendicular to the relative airflow. Lift is generated due to pressure difference between the upper and lower surfaces of the wing.

$$L = \frac{1}{2}\rho V^2 S C_L \tag{1}$$

where:

- ρ is air density,
- V is aircraft velocity,
- S is wing reference area,
- C_L is coefficient of lift.

Lift increases with airspeed, wing area, air density and angle of attack up to the stall limit.

1.1.2. Weight

Weight is the gravitational force acting downward through the aircraft centre of gravity.

$$W = mg \tag{2}$$

where m is aircraft mass and g is acceleration due to gravity.

1.1.3. Thrust

Thrust is the forward force produced by the propulsion system. In piston aircraft it is usually produced by a propeller, while in jet aircraft it is produced by turbine engines.

1.1.4. Drag

Drag is the aerodynamic resistance opposing aircraft motion. It acts opposite to the direction of flight.

$$D = \frac{1}{2}\rho V^2 S C_D \tag{3}$$

where C_D is the coefficient of drag.

Total drag is generally divided into:

- **Parasite drag**: due to skin friction, form drag and interference drag.
- **Induced drag**: produced as a consequence of lift generation.

1.1.5. Condition for Steady Level Flight

For steady, unaccelerated level flight:

$$L = W \tag{4}$$

$$T = D \tag{5}$$

This means the upward and downward forces are balanced, and the forward and backward forces are balanced.

1.2. Aviation Fuels

Aircraft fuel selection depends on engine type, operating temperature, freezing point, flash point, volatility, energy content, safety, availability and environmental impact.

1.2.1. Jet A

Jet A is a kerosene-based aviation turbine fuel mainly used in the United States. It is used in gas turbine engines. It has good energy density, good combustion stability and lower volatility compared to aviation gasoline.

1.2.2. Jet A-1

Jet A-1 is also a kerosene-based turbine fuel and is widely used internationally. The major difference between Jet A and Jet A-1 is that Jet A-1 has a lower freezing point, making it more suitable for international and high-altitude operations.

1.2.3. Avgas

Avgas, or aviation gasoline, is used in spark-ignition piston engines. It has high octane rating to prevent knocking. Common grades include 100LL, where LL stands for low lead. Avgas is not normally used in turbine engines.

1.2.4. Biofuel and Sustainable Aviation Fuel

Biofuel or Sustainable Aviation Fuel, also called SAF, is produced from renewable or waste-derived feedstocks. It is generally blended with conventional jet fuel after qualification and certification. SAF is used to reduce lifecycle carbon emissions while maintaining compatibility with existing turbine engines and fuel infrastructure.

Table 1: Comparison of Aviation Fuels

Fuel	**Main Use**	**Composition**	**Important Properties**	**Selection Criteria**
Jet A	Turbine engines, mostly USA	Kerosene-range hydrocarbons	Good thermal stability, moderate freezing point, low volatility	Used for turbine aircraft operating in moderate climates
Jet A-1	Turbine engines, international aviation	Kerosene-range hydrocarbons with additives	Lower freezing point than Jet A, suitable for high-altitude operations	Preferred for international and cold-weather operations
Avgas	Piston-engine aircraft	Gasoline-range hydrocarbons, octane improvers, sometimes TEL	High octane rating, more volatile than jet fuel	Used only when engine requires spark-ignition gasoline fuel
Biofuel / SAF	Turbine engines, blended with jet fuel	Renewable or waste-derived synthetic hydrocarbons	Lower lifecycle emissions, must meet aviation fuel standards	Selected for sustainability and compatibility with jet engines

1.2.5. Fuel Selection Criteria

The following factors are considered while selecting aviation fuel:

- **Engine compatibility**: turbine engines use jet fuel; piston engines use Avgas.
- **Freezing point**: fuel should not freeze at cruise altitude.
- **Flash point**: higher flash point improves ground handling safety.
- **Energy density**: higher energy per unit mass improves range.
- **Volatility**: excessive volatility can cause vapour lock.
- **Contamination resistance**: water and particles must be controlled.
- **Environmental impact**: SAF is preferred where emission reduction is required.

1.3. Types of Fuel Tanks in Aircraft

Fuel tanks store fuel safely while maintaining aircraft balance and fuel availability during different flight attitudes.

Table 2: Types of Aircraft Fuel Tanks

Tank Type	Description	Advantages	Disadvantages
Bladder Tank	Flexible rubberized tank installed inside a cavity	Leak resistant, replaceable, suitable for small aircraft	Takes up internal space, aging of material possible
Fuselage Tank	Fuel tank located inside fuselage	Easy access, short fuel line length possible	Can affect centre of gravity, safety risk near cabin
Integral / Wing Tank	Aircraft wing structure itself acts as the tank	Saves weight and space, common in transport aircraft	Requires good sealing, leak repair can be difficult

1.4. Parts of an Aircraft

The main parts of an aircraft are shown below.

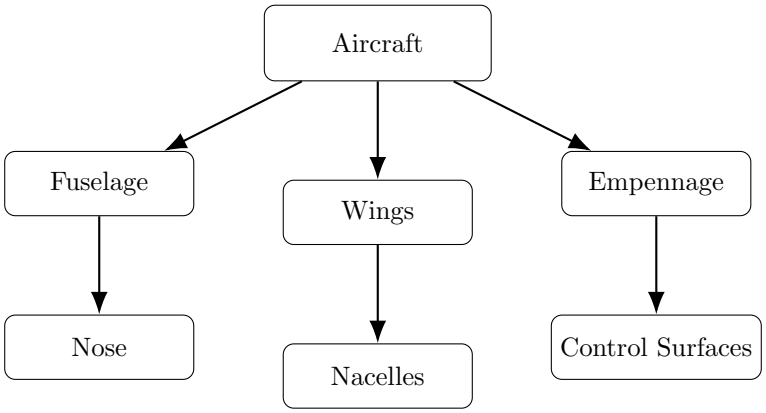


Figure 2: Basic aircraft structural parts

1.4.1. Fuselage

The fuselage is the main body of the aircraft. It carries passengers, cargo, cockpit, avionics and sometimes fuel tanks.

1.4.2. Nose

The nose is the front portion of the aircraft. It usually contains weather radar, avionics equipment and aerodynamic fairing.

1.4.3. Empennage

The empennage is the tail section of the aircraft. It provides stability and control in pitch and yaw.

1.4.4. Horizontal Stabiliser

The horizontal stabiliser provides longitudinal stability. It helps the aircraft resist unwanted pitch motion.

1.4.5. Vertical Stabiliser

The vertical stabiliser provides directional stability. It helps the aircraft resist unwanted yaw motion.

1.4.6. Elevator

The elevator is a movable control surface attached to the horizontal stabiliser. It controls pitch.

1.4.7. Rudder

The rudder is attached to the vertical stabiliser. It controls yaw.

1.4.8. Ailerons

Ailerons are located near the outer trailing edges of the wings. They control roll by moving differentially. When one aileron goes up, the other goes down.

1.4.9. Flaps

Flaps are high-lift devices located on the trailing edge of the wing. They increase lift during takeoff and landing.

1.4.10. Slats

Slats are high-lift devices located on the leading edge of the wing. They delay stall by allowing the wing to operate at higher angle of attack.

1.4.11. Nacelles

Nacelles are streamlined housings for engines. They reduce drag and provide structural support for engine installation.

1.4.12. Stabilator

A stabilator is an all-moving horizontal tail surface. Unlike a normal elevator, the entire stabilator moves to control pitch.

1.4.13. V-Tail

A V-tail uses two angled tail surfaces instead of separate horizontal and vertical stabilisers. These surfaces combine pitch and yaw control.

1.4.14. T-Tail

A T-tail places the horizontal stabiliser on top of the vertical stabiliser.

Table 3: Pros and Cons of Aircraft Tail Configurations

Tail Type	Advantages	Disadvantages
Conventional Tail	Simple, reliable, easy to design and maintain	May be affected by wing wake at high angle of attack
T-Tail	Horizontal tail remains away from wing wake in normal flight, useful for rear-engine aircraft	Higher structural weight, risk of deep stall in some aircraft
V-Tail	Lower number of surfaces, reduced drag and weight possible	Complex control mixing, yaw and pitch coupling
Stabilator	Strong pitch authority, useful at high speed	Sensitive control response, requires anti-servo tab or stability augmentation

1.5. Aircraft Fuel System

The aircraft fuel system stores, filters, transfers and supplies fuel to the engine under all operating conditions.

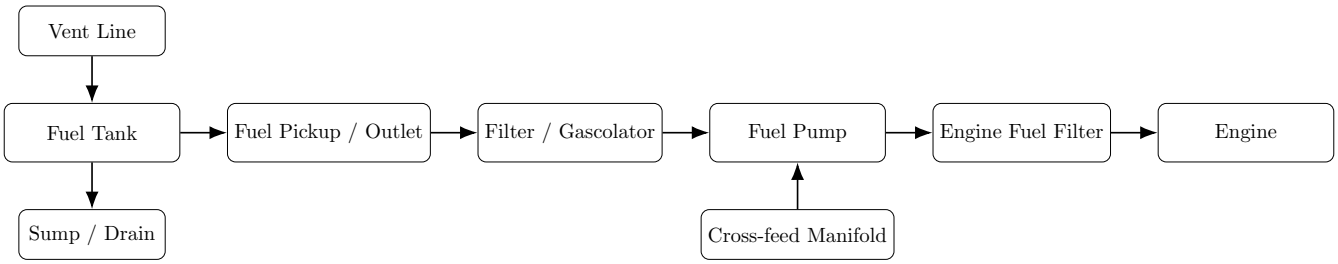


Figure 3: Block Diagram of Aircraft Fuel System

1.5.1. Fuel Pickup / Outlet

The fuel pickup or outlet is the point through which fuel leaves the tank and enters the fuel supply line. It is placed so that fuel can be drawn even during normal aircraft attitude changes.

1.5.2. Sump and Drain

The sump is the lowest portion of the fuel tank or system where water and heavier contaminants collect. The drain allows these contaminants to be removed during inspection.

1.5.3. Vent Lines

Vent lines allow air to enter the fuel tank as fuel is consumed. Without venting, a vacuum would form inside the tank, restricting fuel flow.

1.5.4. One-Way Valves

One-way valves prevent reverse flow of fuel. They help maintain correct flow direction and prevent fuel from flowing back into another tank or line.

1.5.5. Cross-Feed Manifold

A cross-feed manifold allows one engine to receive fuel from another tank. This is useful for fuel balancing, failure management and multi-engine operations.

1.5.6. Flexible Hose Assemblies

Flexible hoses are used where movement, vibration or engine motion is present. They absorb relative movement between components.

1.5.7. Rigid Aluminium and Stainless Steel Tubing

Rigid tubes are used in fixed sections of the aircraft fuel system. Aluminium tubing is lightweight, while stainless steel tubing is used in high-temperature or high-strength areas.

1.6. Filters Used in Aircraft Fuel Systems

Fuel filters remove particles, water and other contaminants before fuel reaches the engine.

Table 4: Aircraft Fuel Filters

Filter Type	Purpose	Remarks
Gascolator	Separates water and solid particles by gravity	Common in light aircraft fuel systems
Mesh Filter	Traps particles using a fine screen	Simple and reusable if cleaned properly
Bypass Filter System	Allows fuel to bypass a blocked filter	Prevents engine starvation, but unfiltered fuel may reach engine
Engine Fuel Filter	Final filtration before fuel enters engine system	Protects fuel injectors, carburettor or fuel control unit

1.7. Fuel Pumps

Fuel pumps ensure that adequate fuel pressure and flow are supplied to the engine.

Table 5: Types of Aircraft Fuel Pumps

Pump Type	Working	Application
Electric Driven Pump	Uses an electric motor to drive the pump	Boost pump, backup pump, starting and priming
Mechanical Driven Pump	Driven by the engine accessory gearbox	Main engine fuel supply in many piston and turbine aircraft
Pneumatic Driven Pump	Driven using air pressure	Used in some specialised aircraft systems

2. Week 2: Aircraft Fire Protection Systems

2.1. Introduction

Aircraft fire protection systems detect, warn and suppress fires in critical zones. These systems are necessary because aircraft contain fuel, electrical systems, hot engines and confined compartments.

The fire triangle consists of:

- fuel,
- heat,
- oxygen.

Fire extinguishing works by removing one or more of these elements.

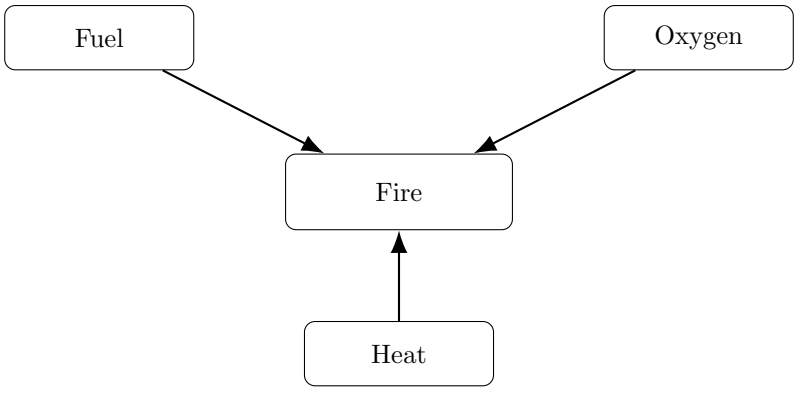


Figure 4: Fire triangle

2.2. Smoke Sensors

Smoke sensors detect smoke particles generated during combustion.

Common aircraft smoke detector types include:

- **Photoelectric smoke detectors**: detect scattered light from smoke particles.
- **Ionisation smoke detectors**: detect changes in ionisation current due to smoke.

Smoke detectors are commonly used in lavatories, cargo compartments and avionics bays.

2.3. Flame Sensors

Flame sensors detect radiation emitted by flames. Aircraft flame detection may use ultraviolet, infrared or combined UV/IR sensing.

2.3.1. Block Diagram of a Flame Sensor

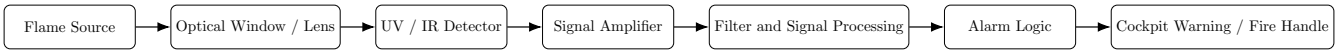


Figure 5: Caption

2.4. Heat Sensors

Heat sensors detect excessive temperature or rapid temperature rise.

Table 6: Types of Heat Sensors

Sensor Type	**Working**	**Application**
Fixed Temperature Sensor	Activates when temperature exceeds a preset value	Engine compartments, APU zones
Rate-of-Rise Sensor	Activates when temperature increases rapidly	Detects sudden fire before absolute temperature becomes very high
Continuous Loop Sensor	Uses a sensing cable routed around fire zones	Common in engines and nacelles

2.5. Handheld Fire Extinguishers

Handheld extinguishers are used by crew to suppress accessible fires inside the aircraft cabin or cockpit.

Table 7: Comparison of Handheld Aircraft Fire Extinguishers

Extinguisher	**Advantages**	**Disadvantages**	**Typical Use**
Halon 1211	Very effective, leaves little residue, suitable for electrical fires	Ozone-depleting agent, being replaced in many applications	Cabin and cockpit fire fighting
Halotron / Halon Replacement	Cleaner agent, lower environmental impact than Halon, suitable for electrical equipment	May require larger quantity than Halon for same effectiveness	Replacement for Halon-based handheld extinguishers
Water-Based Extinguisher	Good cooling effect, useful for paper, cloth and cabin material fires	Not suitable for live electrical or flammable liquid fires	Cabin Class A fires
CO_2 Extinguisher	Leaves no residue, useful for electrical fires	Can displace oxygen, less cooling effect, not ideal in confined cabin spaces	Ground or selected maintenance use

2.6. Aircraft Fire Zones

Aircraft fire zones are areas where fire detection and extinguishing systems are required due to higher fire risk.

Table 8: Fire Zones in Aircraft

Zone	**Fire Risk**	**Protection Used**
Engine Zone	Fuel leakage, hot surfaces, oil leakage	Heat sensors, fire detection loops, extinguishing bottles
APU Zone	Fuel, oil and hot exhaust gases	Fire detection loop and automatic or manual extinguishing
Cargo Zone	Smoke, dangerous goods, electrical faults	Smoke detectors and fixed extinguishing system
Lavatory Zone	Waste bin fire, electrical fault	Smoke detector and automatic waste-bin extinguisher

2.7. Types of Extinguishing Agents

Table 9: Extinguishing Agents

Agent	Working Principle	Remarks
Halogens / Halon	Interrupts chemical chain reaction of fire	Very effective but environmental restrictions apply
CO_2	Displaces oxygen around fire	No residue but oxygen displacement risk exists
Water	Removes heat by cooling	Effective for ordinary combustibles, not live electrical fires
Foam	Smothers fire and separates fuel from oxygen	Useful for liquid fuel fires, not preferred for aircraft electrical areas

2.8. Regulatory Standards for Testing and Maintenance

Aircraft fire protection systems must be inspected, tested and maintained according to applicable airworthiness requirements, manufacturer maintenance manuals and aviation authority regulations.

Important references include:

- FAA Advisory Circular AC 20-42D for aircraft hand fire extinguishers.
- FAA Advisory Circular AC 20-135 for powerplant and APU fire protection compliance guidance.
- EASA CS-25, CS-27 and CS-29 requirements depending on aircraft category.
- Aircraft Maintenance Manual procedures for fire bottle weighing, pressure checks and squib continuity checks.
- Scheduled inspection of smoke detectors, heat detectors, fire loops and warning circuits.
- Hydrostatic testing and pressure inspection of fire extinguisher cylinders.

2.9. Safety Standards

Typical fire protection safety requirements include:

- Fire detection must provide timely warning to the crew.
- Fire extinguishing must be effective for the protected zone.
- Extinguishing agents should not create unacceptable toxicity risk.
- Wiring and sensors must withstand vibration and temperature.
- False warnings should be minimized.
- Maintenance checks must verify sensor continuity and bottle pressure.

3. Week 3: Rotorcraft Avionics, Navigation and Communication Systems

3.1. Unique Requirements of Rotorcraft Avionics

Rotorcraft operate differently from fixed-wing aircraft. Helicopters can hover, fly at low altitude, operate near obstacles and land in confined areas. Therefore, rotorcraft avionics must support low-altitude navigation, obstacle awareness, vibration monitoring and mission-specific communication.

Table 10: Unique Requirements of Rotorcraft Avionics

Requirement	Reason
Low-altitude terrain awareness	Rotorcraft often fly close to ground, terrain, towers and wires
High vibration tolerance	Rotor systems produce vibration affecting sensors and electronics
Hover and low-speed navigation	Helicopters may hover or fly slowly during rescue, offshore and military missions
Obstacle and traffic awareness	Required during EMS, offshore, urban and mountain operations
Health monitoring	Rotors, gearboxes and drive systems require continuous monitoring
Mission integration	EO/IR gimbals, searchlights, hoists and tactical radios may be integrated

3.2. TAWS: Terrain Awareness and Warning System

TAWS warns the pilot when the aircraft is approaching terrain or obstacles dangerously. Helicopter TAWS is especially important because rotorcraft frequently operate at low altitude.

3.2.1. TAWS Architecture

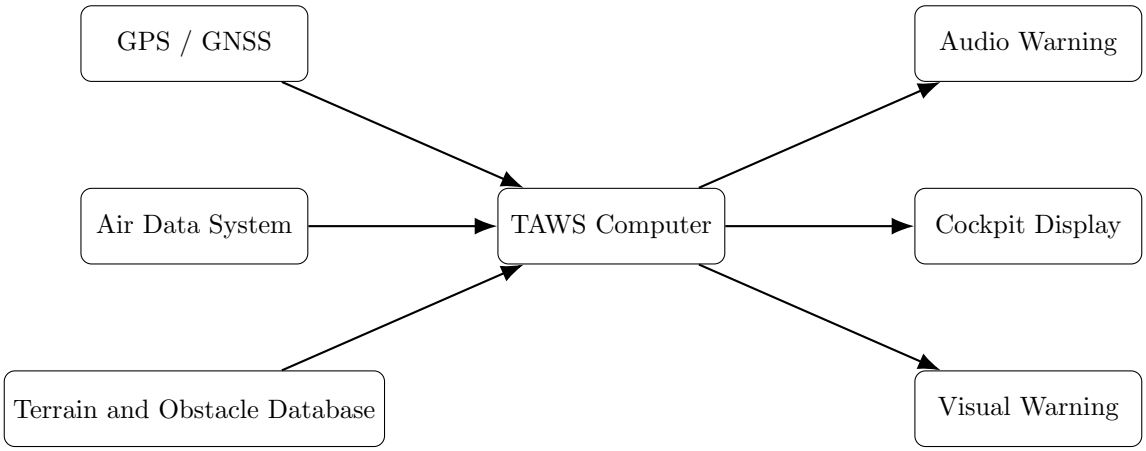


Figure 6: Basic TAWS architecture

3.3. HUMS: Health and Usage Monitoring System

HUMS monitors the condition of critical rotorcraft components such as rotor blades, gearbox, shafts, bearings and engines. It helps detect vibration signatures, imbalance, wear and early faults.

3.3.1. HUMS Architecture



Figure 7: Caption

3.4. VHF, VOR, VOR/DME, VOR Navigation and VORTAC

Table 11: Radio Navigation Systems

System	Use	Brief Working	Where Used
VHF Communication	Voice communication	Uses VHF radio band for pilot-ATC communication	Civil and military aviation
VOR	Bearing navigation	Ground station sends reference and variable phase signals; aircraft determines radial	En-route and terminal navigation
VOR/DME	Bearing plus distance	VOR gives direction, DME gives slant range distance	Airways, approaches and position fixing
VOR Navigation	Navigation using VOR radials	Pilot tracks selected radial to or from station	Conventional IFR navigation
VORTAC / TACAN	Civil VOR combined with military TACAN	Provides bearing and distance information	Civil-military shared navigation sites

3.5. JPALS

JPALS stands for Joint Precision Approach and Landing System. It is a precision landing system that uses differential GPS corrections and secure data links to guide aircraft during approach and landing. It is especially useful for shipboard and military operations where conventional landing aids may be unavailable or unsuitable.

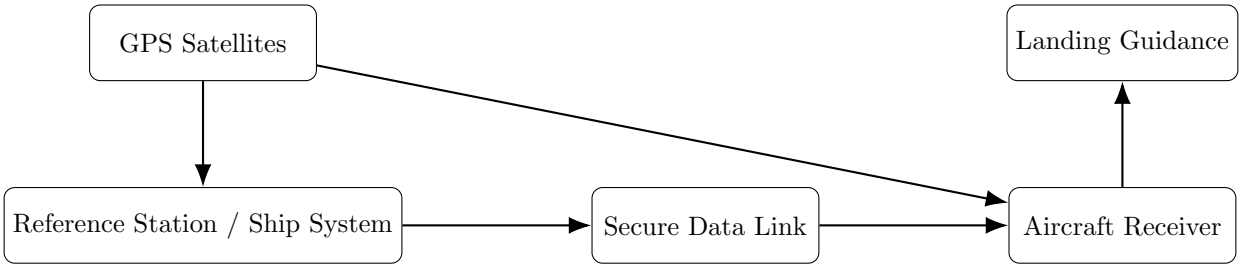


Figure 8: Simplified JPALS architecture

3.6. ARA, PinS, RNAV/RNP and Doppler-Based Navigation

Table 12: Rotorcraft Navigation Systems: Working, Use, Pros and Cons

System	Use	Brief Working	Advantages	Limitations
JPALS	Precision landing	Uses differential GPS correction and data link guidance	High precision, useful for shipboard landing	Requires compatible equipment and data link
ARA	Airborne radar approach / aircraft radar approach	Uses onboard radar returns to assist approach in low visibility or offshore areas	Useful where ground aids are limited	Depends on radar interpretation and environment
PinS	Point-in-Space approach	GNSS-based approach to a point near heliport followed by visual segment	Useful for helicopter operations in obstacle-rich areas	Final visual segment may depend on weather minima
RNAV	Area navigation	Allows aircraft to fly any desired path using GNSS, DME/DME or INS	Flexible routing, reduces dependence on ground beacons	Accuracy depends on sensor availability
RNP	Required Navigation Performance	RNAV with onboard monitoring and alerting	Higher navigation assurance, useful in complex terrain	Requires certified aircraft and crew approval
Doppler Navigation	Velocity and drift measurement	Measures frequency shift between transmitted and received signals	Useful when GPS is unavailable, gives ground speed and drift	Accuracy affected by surface reflection and sensor errors

3.6.1. Doppler Shift Equation

Doppler navigation uses the change in frequency caused by relative motion.

$$f_d = \frac{2v}{\lambda} \cos\theta \tag{6}$$

where:

- f_d is Doppler frequency shift,
- v is aircraft velocity relative to the ground,
- λ is wavelength,
- θ is angle between beam direction and velocity vector.

The aircraft velocity can be estimated from:

$$v = \frac{f_d \lambda}{2\cos\theta} \tag{7}$$

3.7. Communication Systems in Rotorcraft

Rotorcraft communication systems include voice radios, data links, satellite communication, emergency beacons and mission communication equipment.

Table 13: Rotorcraft Communication Systems

System	Purpose	Remarks
VHF Radio	Pilot-to-ATC voice communication	Main civil aviation communication system
UHF / Military Radio	Tactical and military communication	Used in military rotorcraft
SATCOM	Beyond line-of-sight communication	Useful for offshore, rescue and remote operations
Modem-Based Rotorcraft Data Link	Data transmission coordinated with rotor blade movement	Reduces communication blockage or interference caused by rotating blades
Gateways	Connect different communication networks	Useful in rescue, military and offshore operations
EO/IR Gimbal Data Link	Transfers video or targeting data from gimballed camera systems	Used in surveillance, search and rescue, police and defence helicopters
Proximity Traffic Alerts	Warns about nearby aircraft	Improves traffic awareness in low-altitude airspace
Low-Altitude ADS-B	Broadcasts aircraft position and receives traffic information	Useful for situational awareness near terrain and heliports
Automatic Direction Finder	Receives NDB signals and indicates bearing to station	Older but still useful as backup navigation aid

3.8. Automatic Direction Finder

ADF receives signals from a Non-Directional Beacon. It indicates the bearing of the station relative to the aircraft nose. It is simple and useful for backup navigation, but it is affected by weather, terrain and night effect.

4. Week 4: Aircraft Electrical Systems

4.1. Introduction

Aircraft electrical systems generate, convert, distribute and manage electrical power. Electrical power is required for avionics, lighting, communication, navigation, flight control computers, pumps, fans, anti-icing systems and emergency systems.

4.2. DC and AC Power Systems

Aircraft may use both DC and AC systems.

Table 14: Aircraft AC and DC Electrical Systems

System	Typical Voltage	Loads
DC System	14 V DC, 28 V DC or higher-voltage DC in modern aircraft	Battery charging, avionics, lighting, relays, emergency loads
AC System	115 V AC, 400 Hz three-phase in many transport aircraft	Motors, pumps, galley loads, transformer-rectifier units, large aircraft systems

4.3. Power Conversion

Aircraft use power conversion devices to supply different loads.

4.3.1. Rectifier

A rectifier converts AC to DC.

$$V_{DC} \approx \frac{2V_m}{\pi} \tag{8}$$

for a single-phase full-wave rectifier, where V_m is peak input voltage.

4.3.2. Inverter

An inverter converts DC to AC. In aircraft, inverters may be used to supply AC loads from a DC bus or battery.

4.3.3. Transformer

A transformer changes AC voltage level.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \tag{9}$$

where:

- V_s is secondary voltage,
- V_p is primary voltage,
- N_s is secondary turns,
- N_p is primary turns.

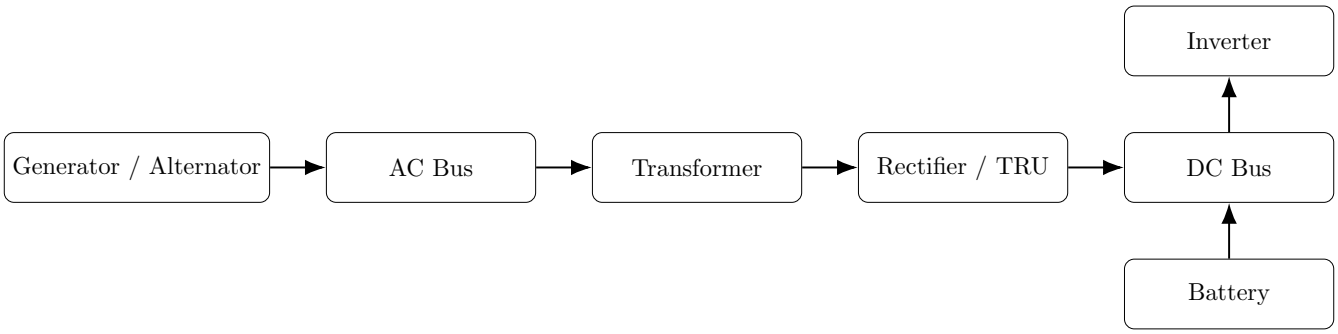


Figure 9: Basic Electrical Power Conversion in Aircrafts

4.4. Generator and Alternator

A generator or alternator converts mechanical energy from the engine into electrical energy.

$$P_{elec} = VI \tag{10}$$

$$P_{mech} = T\omega \tag{11}$$

The efficiency is:

$$\eta = \frac{P_{elec}}{P_{mech}} \tag{12}$$

4.4.1. Generator

A DC generator directly produces DC output using a commutator.

4.4.2. Alternator

An alternator produces AC output. The AC output may be rectified to DC if required. Alternators are common because they are efficient, reliable and suitable for higher power levels.

Table 15: Generator vs Alternator

Parameter	**Generator**	**Alternator**
Output	DC or AC depending on construction	AC output
Main Parts	Armature, field winding, commutator or slip rings	Rotor field, stator winding, rectifier if DC output needed
Efficiency	Lower in older DC systems	Generally higher
Aircraft Use	Older or smaller aircraft systems	Common in modern aircraft electrical systems

4.5. Power Distribution Buses

Aircraft electrical buses distribute power to different loads.

Table 16: Aircraft Electrical Buses

Bus	Purpose	Typical Loads
Main Bus	Supplies normal aircraft loads during normal operation	Lighting, cabin systems, pumps, avionics
Essential Bus	Supplies systems needed for safe continuation of flight	Primary flight instruments, radios, navigation, engine controls
Emergency Bus	Supplies minimum critical systems during major power failure	Standby instruments, emergency lighting, selected communication
Battery Bus	Directly connected or controlled from battery	Emergency loads, starting circuits, fire extinguishing squibs
Non-Essential Bus	Loads that can be removed during failure	Galley, entertainment, comfort systems

4.6. Priority Order During Engine or Alternator Failure

During generator or alternator failure, the aircraft electrical load management system disconnects loads in priority order so that critical systems remain powered.

1. Disconnect galley, cabin entertainment and comfort loads.
2. Disconnect non-essential lighting and service loads.
3. Disconnect secondary pumps and non-critical heaters.
4. Maintain avionics, navigation and communication loads.
5. Maintain flight instruments, engine controls and emergency systems.
6. Use battery, APU generator, standby alternator or ram air turbine if available.

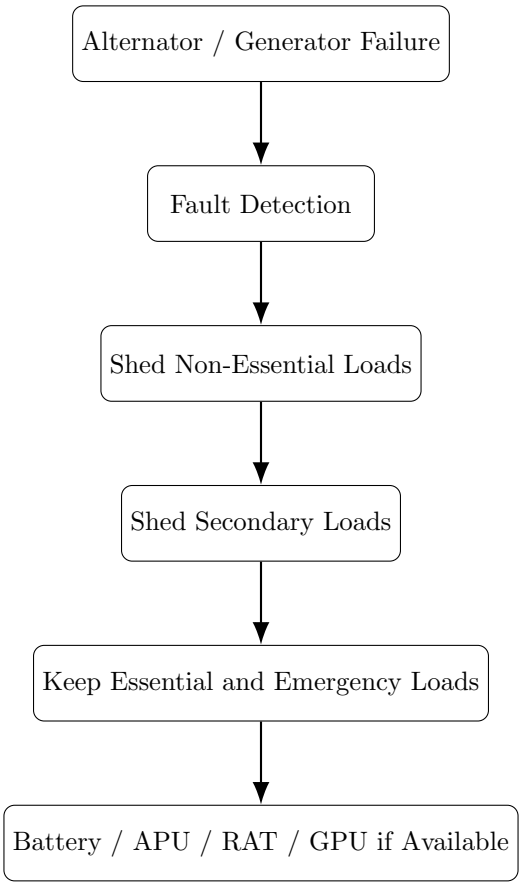


Figure 10: Load shedding priority during electrical failure

4.7. Electrical Load Management System

The Electrical Load Management System monitors generation capacity, bus voltage, current and system faults. It automatically disconnects low-priority loads when available power is insufficient.

4.8. Ground Power Unit Architecture

A Ground Power Unit, or GPU, supplies electrical power to an aircraft while it is on the ground. This reduces the need to run the aircraft engines or APU.

Common GPU outputs include:

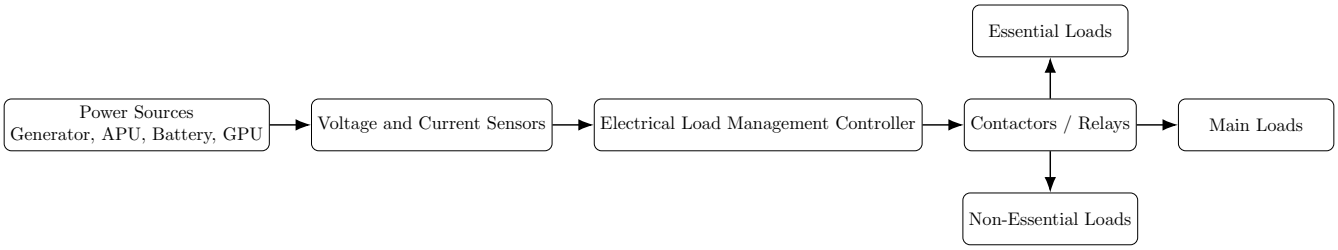


Figure 11: Electrical Load Management System

- 115 V AC, 400 Hz, three-phase supply for large aircraft.
- 28 V DC supply for smaller aircraft or DC systems.

4.8.1. GPU Block Diagram



Figure 12: Ground Power Unit Architecture and External Connection

4.9. External Power Receptacle

Large transport aircraft such as the Boeing 737 use an external power receptacle to connect the GPU to the aircraft electrical system. The receptacle allows safe connection of external AC power. After connection, the aircraft electrical system checks phase sequence, voltage, frequency and protection interlocks before connecting external power to the bus.

4.10. Essential and Non-Essential AC and DC Consumers

Table 17: Essential and Non-Essential Electrical Consumers

Category	**AC Consumers**	**DC Consumers**
Essential	Flight control computers, critical pumps, navigation systems, anti-ice equipment depending on aircraft type	Standby instruments, communication radios, emergency lights, engine control units
Non-Essential	Galley loads, cabin comfort loads, passenger entertainment, non-critical fans	Cabin lighting, service outlets, non-critical relays and indicators
Emergency	Standby AC instruments through inverter	Battery-powered instruments, emergency lighting, fire extinguishing circuits

4.11. Pros and Cons of Aircraft Electrical System Elements

Table 18: Pros and Cons of Electrical System Elements

Element	**Advantages**	**Disadvantages**
AC System	Efficient for high power, transformer voltage conversion possible	Requires frequency control and phase management
DC System	Easy battery integration, suitable for electronics	High current at low voltage causes conductor losses
Battery	Emergency backup power, engine/APU starting support	Limited energy capacity, requires charging and maintenance
GPU	Saves fuel on ground, reduces APU usage	Requires ground infrastructure and proper connection checks
Load Shedding System	Protects essential loads during failure	Incorrect prioritisation can disconnect useful systems
Transformer Rectifier Unit	Converts AC to DC for DC buses	Adds weight, heat and failure points

5. Week 5:Design and Simulation of a Ground Power Unit

5.1. Objective

The objective of this section is to design and simulate an aircraft Ground Power Unit (GPU) capable of providing regulated aircraft-compatible power supplies. The target outputs are:

- Three-phase AC output of 115V RMS phase-to-phase.
- Three-phase AC output of approximately 200V RMS line-to-line.
- AC output frequency of 400Hz.
- Regulated 28V DC output.

The Ground Power Unit is intended to supply power to aircraft electrical loads when the aircraft is on the ground. The 400Hz AC supply is commonly used in aircraft systems because higher frequency operation allows reduction in transformer and filter size. The 28V DC supply is used for low-voltage aircraft loads, avionics, control circuits, and auxiliary systems.

5.2. System Architecture

The proposed Ground Power Unit consists of the following major stages:

1. Three-phase AC input source.
2. Three-phase diode rectifier.
3. DC-link capacitor.
4. Three-phase voltage source inverter.
5. LC output filter.
6. Buck converter for 28V DC generation.
7. AC and DC loads.

The overall power conversion structure is:

$$\text{3-phase AC Input} \rightarrow \text{Diode Rectifier} \rightarrow \text{DC Link} \rightarrow \begin{cases} \text{VSI + LC Filter} \rightarrow 115V/200V, 400Hz \text{ AC} \\ \text{Buck Converter} \rightarrow 28V \text{ DC} \end{cases} \tag{13}$$

The rectifier converts the input AC supply into DC. The DC-link capacitor smoothens the rectified voltage and supplies power to the inverter and buck converter. The inverter generates the 400Hz AC waveform, while the LC filter removes switching harmonics. The buck converter steps down the DC-link voltage to obtain the required 28V DC output.

5.3. Schematic Diagram

Figure 13 shows the simulated schematic of the Ground Power Unit. The schematic includes the rectifier stage, DC-link capacitor, inverter section, output LC filter, and buck converter section.

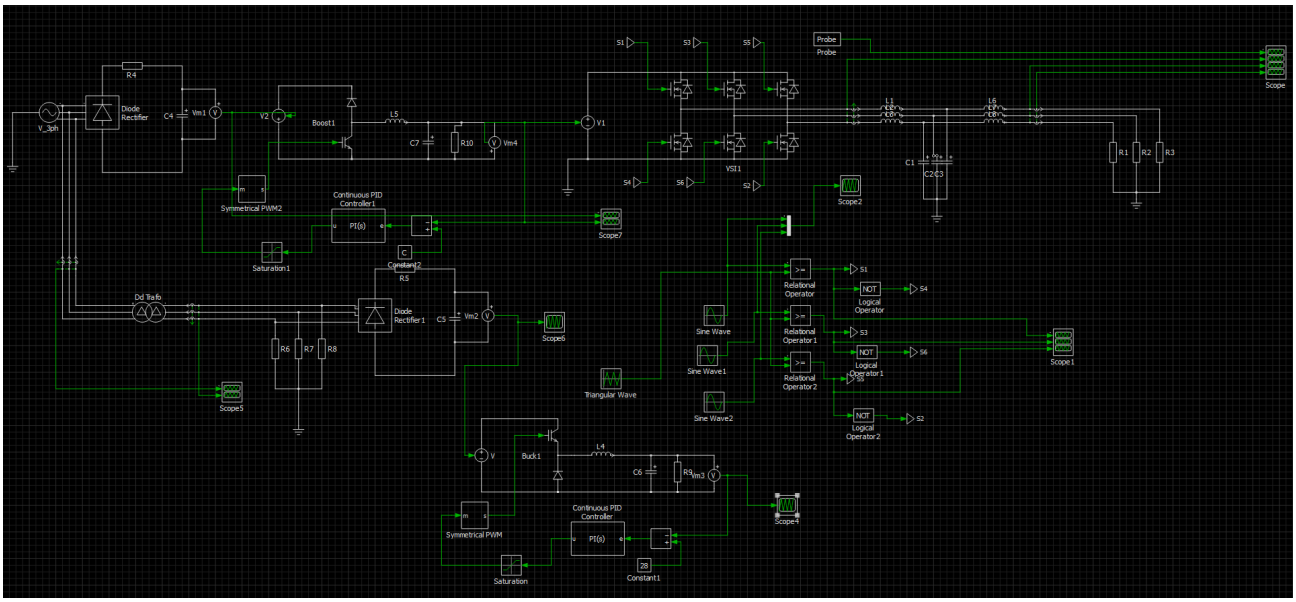


Figure 13: Simulated schematic of the Ground Power Unit

5.4. Design Specifications

The main design specifications of the Ground Power Unit are listed in Table 19.

Table 19: Ground Power Unit Design Specifications

Parameter	Target Value	Remarks
AC phase voltage	115V RMS	Phase-to-neutral voltage
AC line voltage	200V RMS	Line-to-line voltage
AC output frequency	400Hz	Aircraft AC supply frequency
DC output voltage	28V	Aircraft DC supply
Voltage THD	Less than 5%	Desired output quality
DC voltage ripple	Low	Depends on output filter
Input stage	3-phase diode rectifier	AC to DC conversion
DC-DC stage(115V AC)	Boost converter	Step-up conversion
DC-DC stage(28V DC)	Buck converter	Step-down conversion

For a balanced three-phase system, the relation between line voltage and phase voltage is:

$$V_{LL} = \sqrt{3}V_{ph} \tag{14}$$

Therefore, for a phase voltage of 115V RMS:

$$V_{LL} = \sqrt{3} \times 115 \approx 200VRMS \tag{15}$$

Hence, 115V RMS phase-to-neutral and 200V RMS line-to-line represent the same three-phase AC voltage level.

5.5. Rectifier and DC-Link Stage

The three-phase input AC voltage is converted into DC using a three-phase diode rectifier. The approximate average DC output voltage of a three-phase diode rectifier is:

$$V_{dc} \approx 1.35V_{LL,rms} \tag{16}$$

where $V_{LL,rms}$ is the input line-to-line RMS voltage.
The DC-link capacitor is used to reduce the ripple in the rectified DC voltage. The DC-link voltage ripple is calculated as:

$$\Delta V_{dc} = V_{dc,max} - V_{dc,min} \tag{17}$$

The quantities to be recorded at this stage are:

- Input three-phase voltage.
- Input current waveform.
- Rectifier output voltage.
- DC-link voltage.
- DC-link ripple voltage.
- Input current THD.

5.6. Three-Phase Inverter Stage

A three-phase voltage source inverter is used to generate the required 400Hz AC output. Sinusoidal pulse width modulation is used to produce three-phase output voltages. The three reference signals are displaced by 120°:

$$v_a = V_m \sin(2\pi ft) \tag{18}$$
$$v_b = V_m \sin(2\pi ft - 120°) \tag{19}$$
$$v_c = V_m \sin(2\pi ft + 120°) \tag{20}$$

where the required output frequency is:

$$f = 400Hz \tag{21}$$

The inverter output is initially a PWM waveform. Therefore, an LC filter is used to obtain a near-sinusoidal output voltage.

5.7. AC Output Filter

The LC filter reduces the switching harmonics present in the inverter output voltage. The cutoff frequency of the LC filter is given by:

$$f_c = \frac{1}{2\pi\sqrt{LC}} \tag{22}$$

The cutoff frequency should be selected such that:

$$f_{out} < f_c < f_{sw} \tag{23}$$

where f_{out} is the fundamental output frequency and f_{sw} is the inverter switching frequency. Since the required output frequency is 400Hz, the filter cutoff frequency should be higher than 400Hz but lower than the switching frequency.

5.8. Buck Converter for 28 V DC Output

A buck converter is used to generate the regulated 28V DC supply from the DC-link voltage. The ideal voltage conversion ratio of a buck converter is:

$$V_o = DV_{in} \tag{24}$$

where D is the duty cycle, V_{in} is the buck input voltage, and V_o is the output voltage.
Therefore, the required duty cycle is:

$$D = \frac{V_o}{V_{in}} \tag{25}$$

For example, if the buck converter input voltage is 280V, then:

$$D = \frac{28}{280} = 0.1 \tag{26}$$

The main quantities to be observed in the buck converter are:

- Buck input voltage.
- Buck output voltage.
- Duty cycle.
- Inductor current.
- Inductor current ripple.
- Output voltage ripple.
- Output current.

5.9. Total Harmonic Distortion

Total Harmonic Distortion is used to evaluate the quality of the AC output waveform. The voltage THD is calculated as:

$$THD_v = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots + V_n^2}}{V_1} \times 100 \tag{27}$$

where V_1 is the fundamental RMS voltage and $V_2, V_3, \ldots, V_n$ are the harmonic RMS voltage components.
Similarly, the current THD is:

$$THD_i = \frac{\sqrt{I_2^2 + I_3^2 + I_4^2 + \cdots + I_n^2}}{I_1} \times 100 \tag{28}$$

The THD values should be recorded for:

- AC output phase voltage.
- AC output line voltage.
- AC output current.
- Rectifier input current.

5.10. Simulation Parameters

The simulation parameters used in the Ground Power Unit model are listed in Table 20.

Table 20: Simulation Parameters

Parameter	Value Used
Input AC voltage	230V RMS
Input frequency	50Hz
DC-link capacitor	50e-6
Inverter switching frequency	10kHz
AC output frequency	400Hz
AC filter inductance	0.2mH
AC filter capacitance	1uF
Buck switching frequency	25kHz
28V Output Buck inductor	1mH
28V Output Buck capacitor	200uF
AC load resistance	100
DC load resistance	1

5.10.1. AC Output Results

Table 21: AC Output Results

Parameter	Expected Value	Actual Value	Remarks
Phase voltage V_{an}	115V RMS	115.957V RMS	Phase-to-neutral
Phase voltage V_{bn}	115V RMS	117.443V RMS	Phase-to-neutral
Phase voltage V_{cn}	115V RMS	117.749V RMS	Phase-to-neutral
Line voltage V_{ab}	200V RMS	200.843V RMS	Line-to-line
Line voltage V_{bc}	200V RMS	200.636V RMS	Line-to-line
Line voltage V_{ca}	200V RMS	203.948V RMS	Line-to-line
Output frequency	400Hz	SI400Hz	Aircraft AC supply
Voltage THD	$< 5\%$	0.8%	From FFT analysis
Current THD	$< 5\%$	$< 5\%$	From FFT analysis

5.10.2. 28 V DC Output Results

Table 22: 28 V DC Output Results

Parameter	Expected Value	Actual Value	Remarks
Buck input voltage	56	57.0582	DC-link voltage
Buck duty cycle	0.5	0.55	$D = V_o/V_{in}$
DC output voltage	28V	27.8729V	Regulated DC output
Output voltage ripple	Low	0.1167V	Peak-to-peak ripple

5.11. Simulation Waveforms

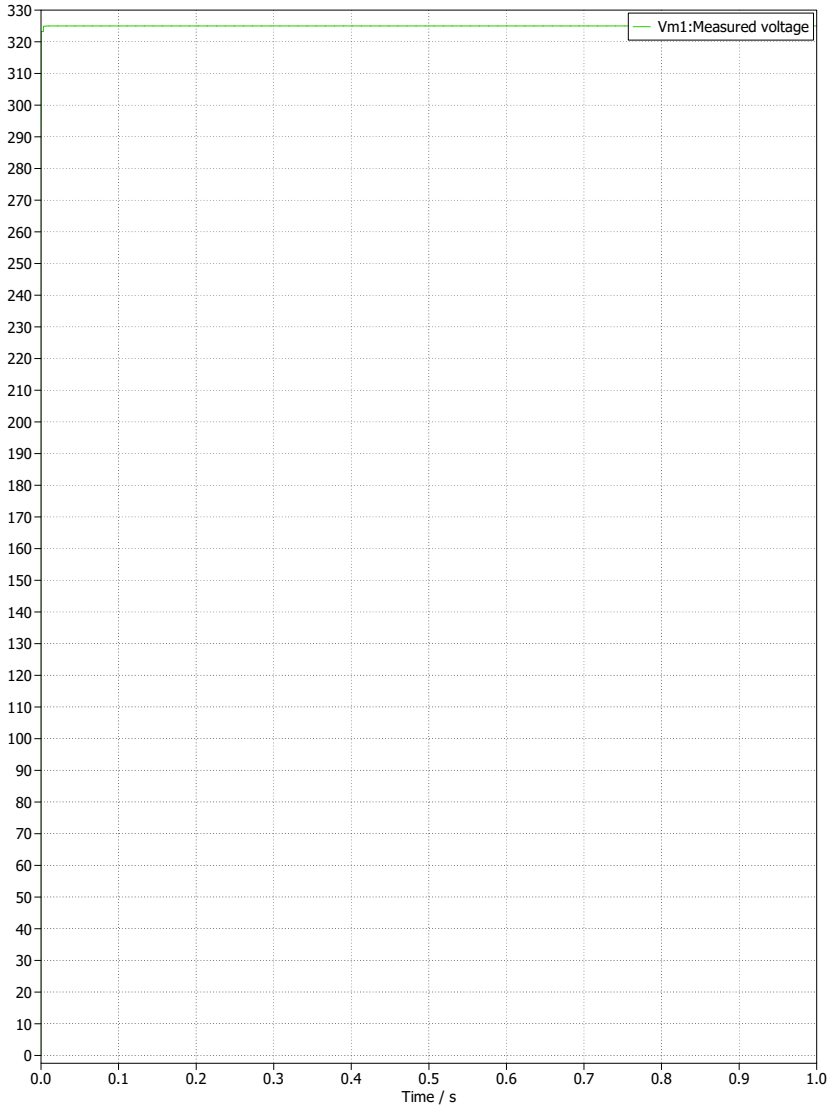


Figure 14: Rectifier output and Boost Converter Output

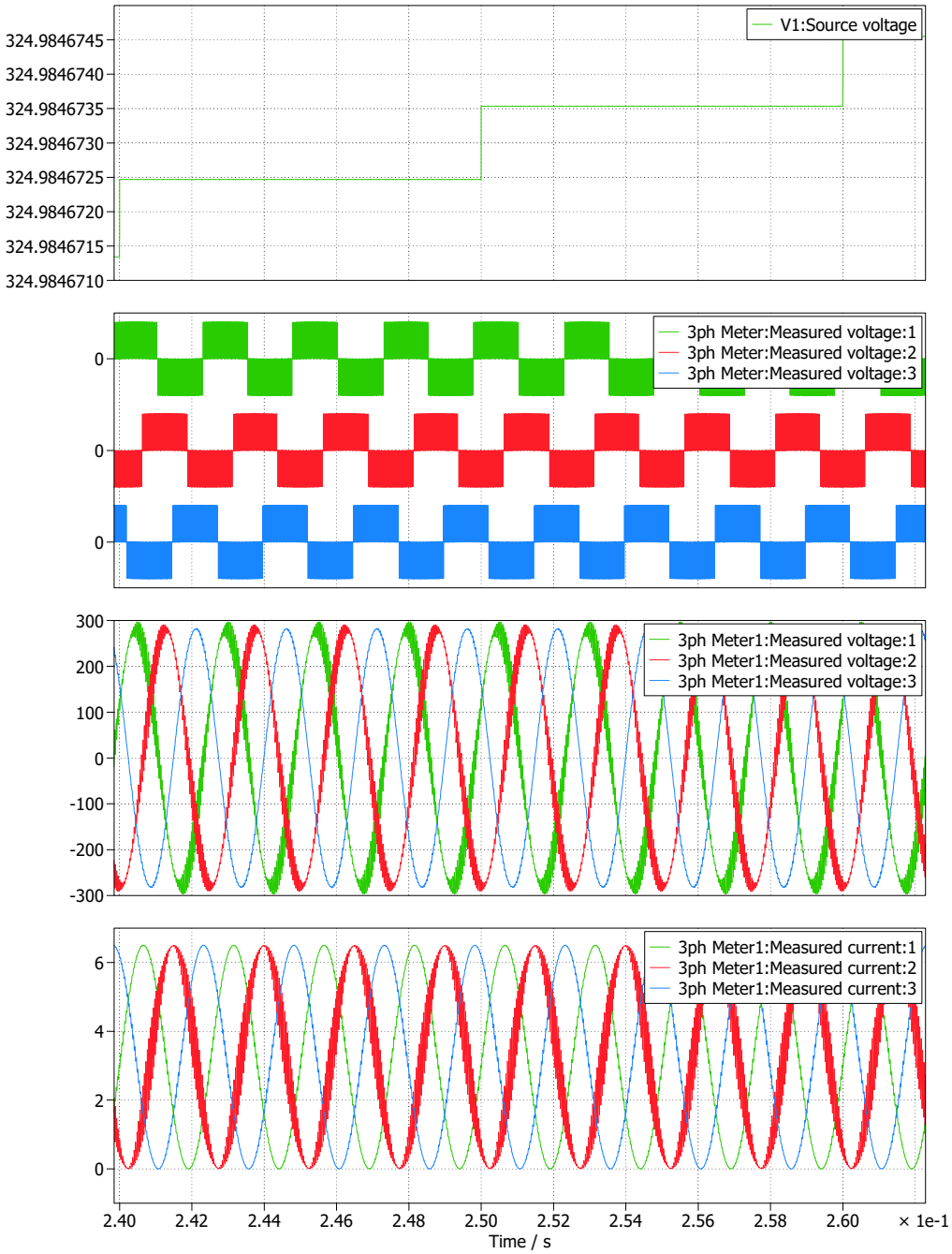


Figure 15: 3 Phase Inverter Outputs

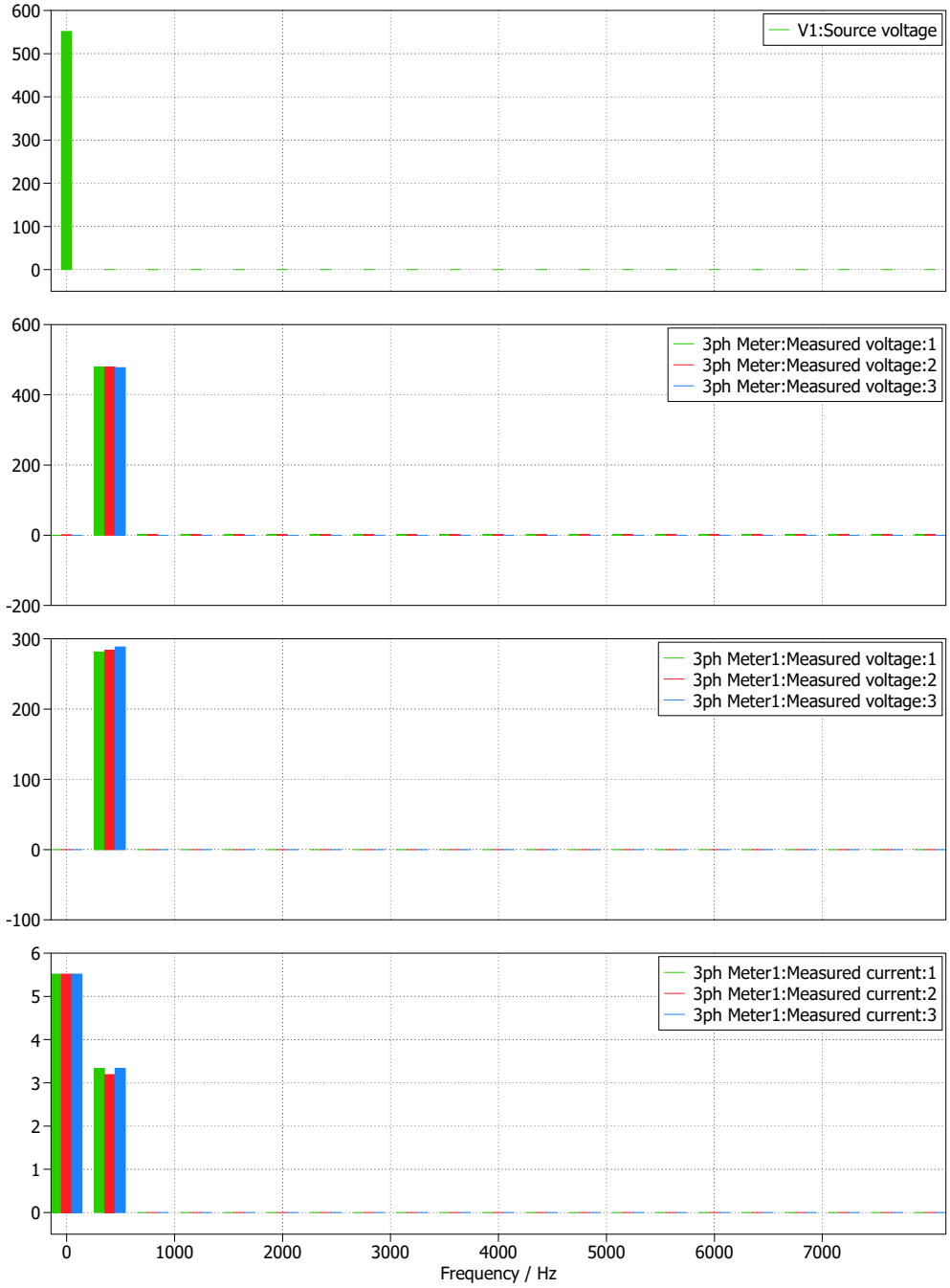


Figure 16: Buck Converter output producing 195V

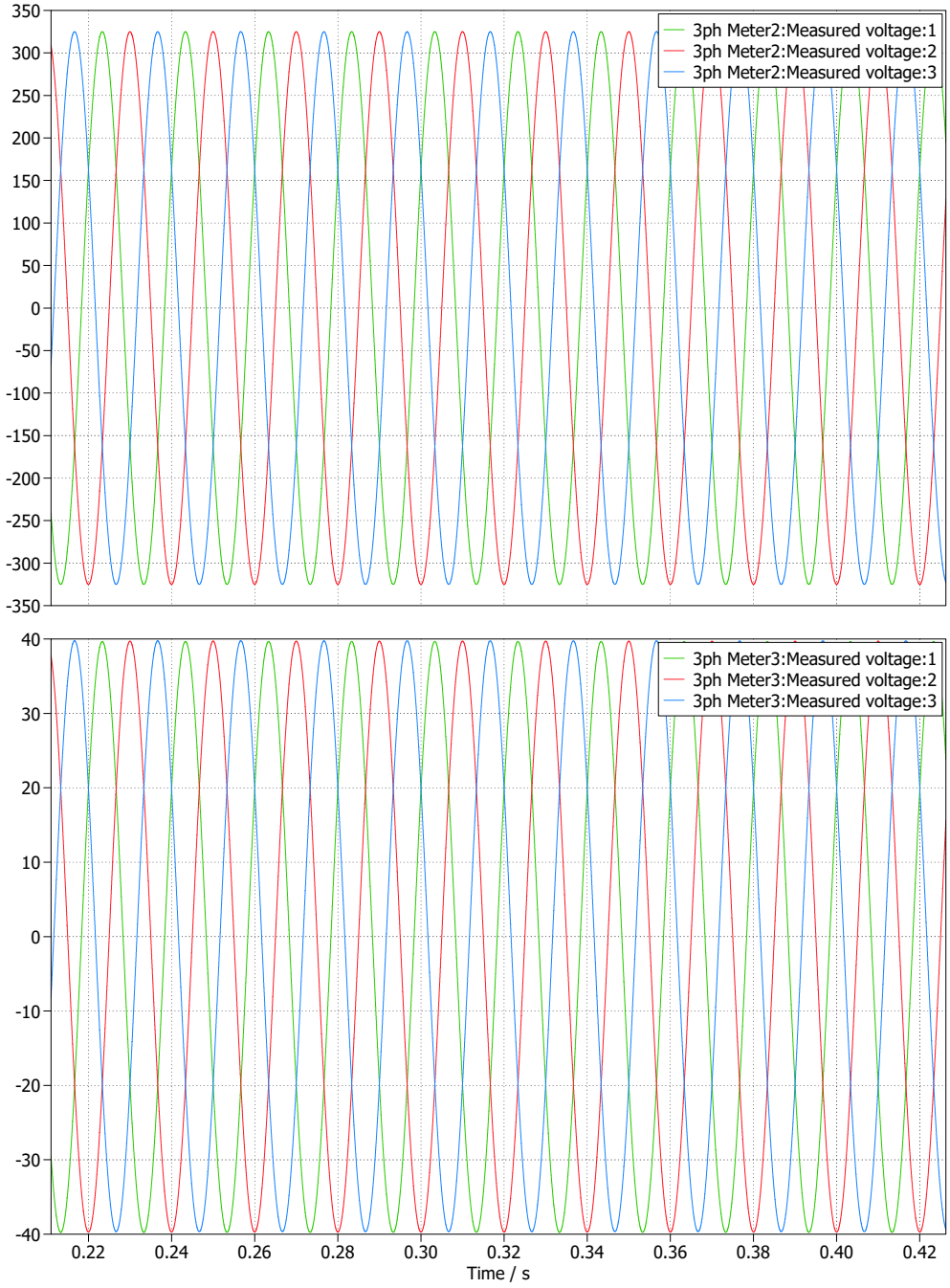


Figure 17: Grid Voltage and Transformer Output to get 28V DC

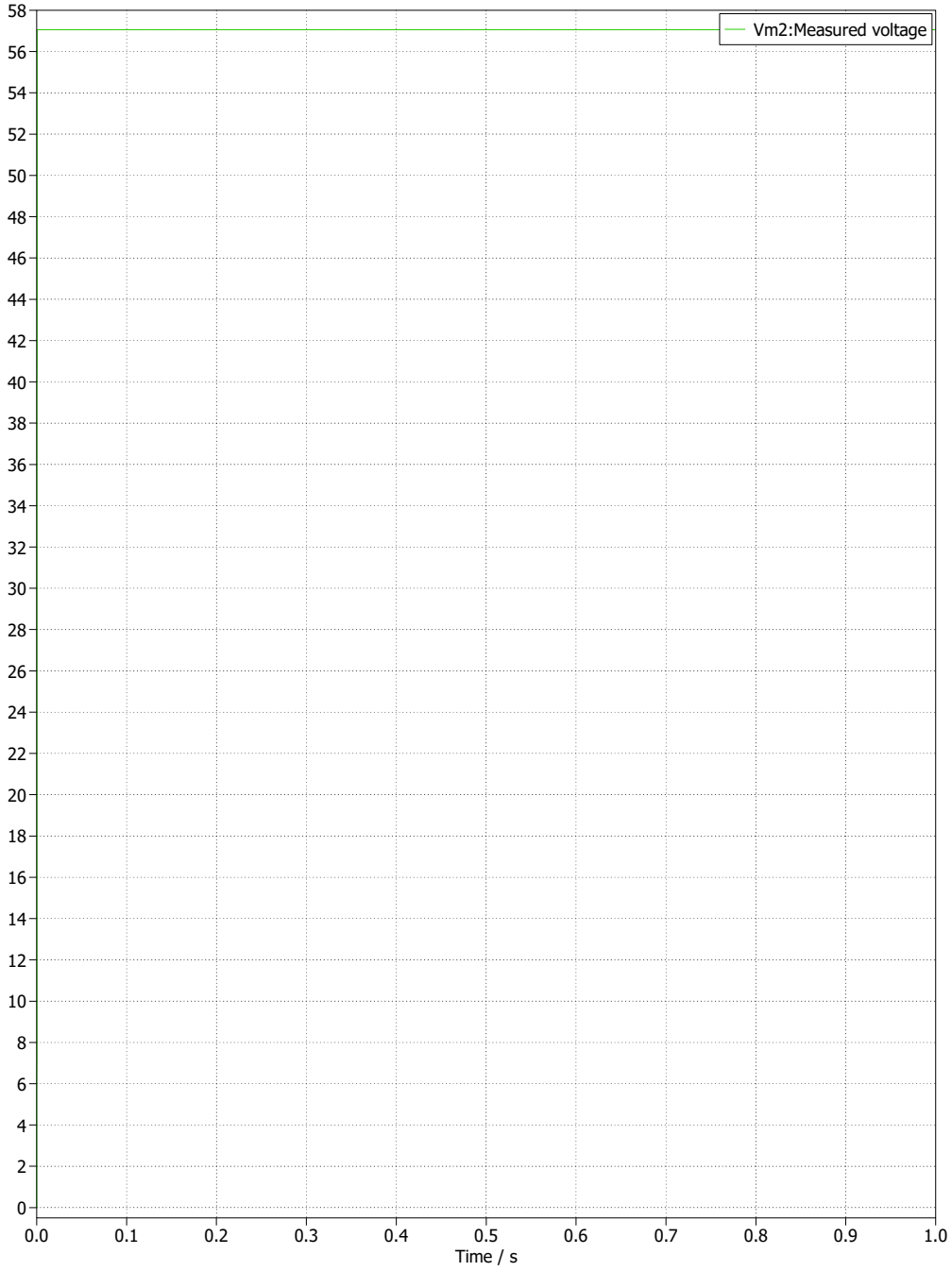


Figure 18: 3 phase rectifier output producing 56V

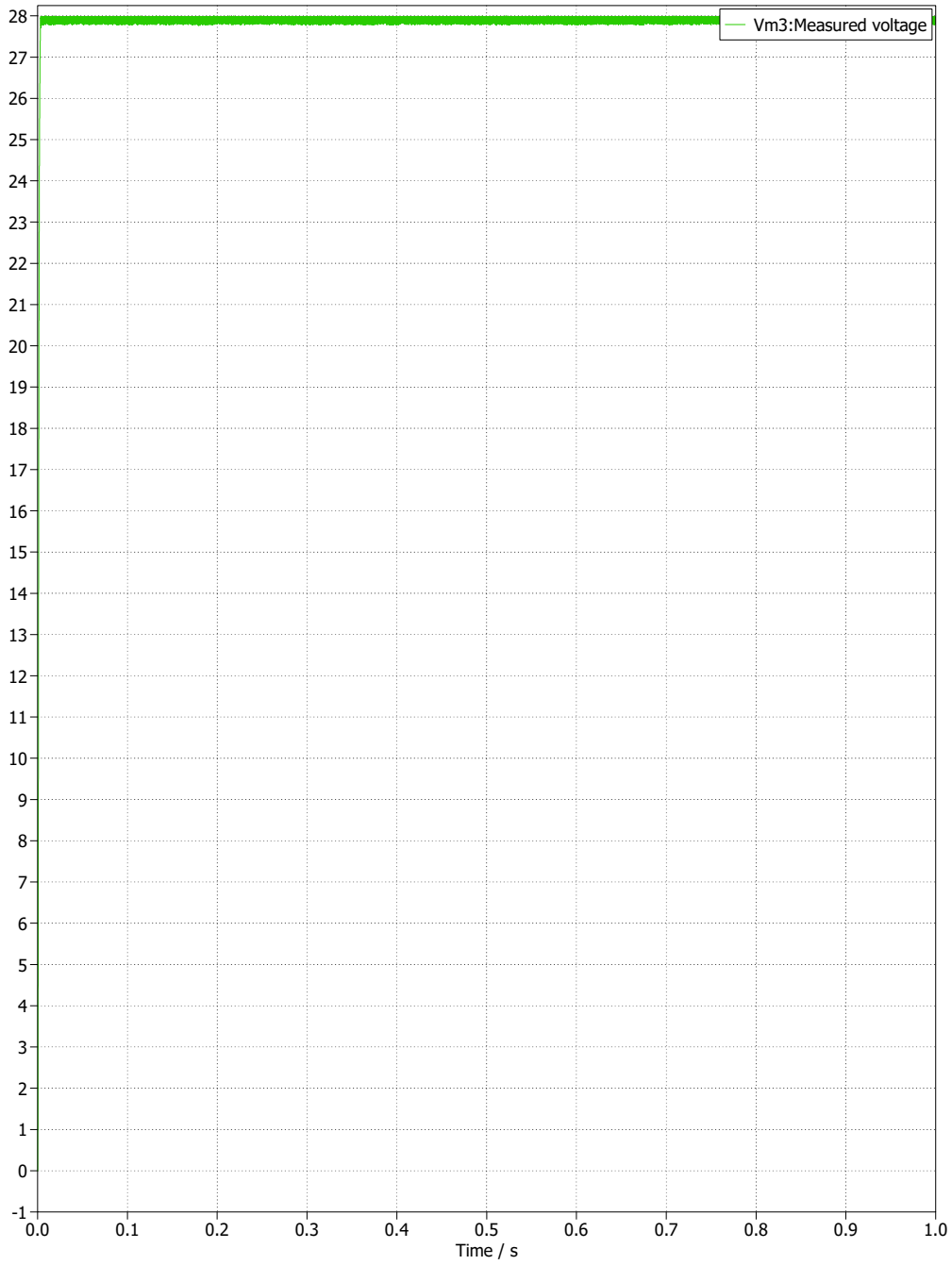


Figure 19: Buck Converter output producing 28V DC

5.12. Expected and Actual Results

Table 23 compares the expected and actual simulation results.

Table 23: Expected and Actual Results

Parameter	Expected	Actual	Error	Remarks
AC phase voltage	115V RMS	114.275V RMS	0.725V	Line Voltage
AC line voltage	200V RMS	200V RMS	0	Line-to-line
AC output frequency	400Hz	400Hz	0	Frequency accuracy
DC output voltage	28V	27.872V	0.128V	Buck output
Grid Side DC-link voltage	325V RMS	325.002V RMS	0.002V	Rectifier output
DC-link voltage(28V)	56	57.0612	1.0612V	Rectifier output
115 V AC voltage THD	$< 5\%$	0.09	0.04	Output voltage quality
DC voltage ripple	Low	0.1172V	0.1072V	Peak-to-peak ripple

5.13. Discussion

The rectifier stage converts the three-phase AC input into an unregulated DC voltage. The DC-link capacitor reduces the ripple in the rectified output and provides a common DC bus for the inverter stage.

The voltage source inverter converts the DC-link voltage into a three-phase AC output. The inverter output is filtered using an LC filter to obtain a near-sinusoidal 400Hz waveform. The AC output voltage is verified using both phase-to-neutral and line-to-line measurements. The expected values are 115V RMS phase-to-neutral and approximately 200V RMS line-to-line.

A three phase transformer delta-delta configuration is connected from the grid to step down the voltage and that is connected to another 3 phase rectifier.The output of this 3 phase rectifier is connected to a capacitor filter to suppress the ripple.the output of this capacitor filter is connected to a buck converter that steps down the DC-link voltage to obtain the required 28V DC supply. The output voltage, output current, duty cycle, inductor current ripple, and voltage ripple are observed to verify the performance of the DC supply.

The THD of the AC output voltage and current is used to evaluate waveform quality. A lower THD indicates better filtering and improved sinusoidal output. If the THD is high, the LC filter values, inverter switching frequency, or modulation index must be adjusted.

5.14. Conclusion

A Ground Power Unit was designed and simulated to obtain aircraft-compatible power supplies. The target AC output was 115V RMS phase-to-neutral or approximately 200V RMS line-to-line at 400Hz. A buck converter was used to generate the required 28V DC output.

The important results to be recorded are the DC-link voltage, inverter output voltage, filtered AC output voltage, line-to-line voltage, output frequency, 28V DC output voltage, output ripple, and THD levels. The expected outputs are 115V RMS phase voltage, 200V RMS line voltage, 400Hz AC frequency, and 28V DC voltage.

The simulation verifies the basic operation of the Ground Power Unit. Any deviation between expected and actual results may be due to incorrect duty cycle, insufficient filtering, improper grounding, high ripple, poor load selection, or switching-related numerical issues.

6. Week 6:Simulation of Power Conversion systems in Aircrafts including APU

7. Conclusion

During the four-week study, the fundamentals of aircraft systems were covered in a structured manner. Week 1 focused on aircraft aerodynamics, fuel types, fuel tanks, aircraft parts and fuel system components. Week 2 covered aircraft fire protection systems including sensors, extinguishing agents, fire zones and safety regulations. Week 3 focused on rotorcraft avionics, navigation systems, communication systems, TAWS and HUMS. Week 4 covered aircraft electrical power systems, AC and DC buses, power conversion, generators, alternators, load shedding and ground power unit architecture.

The study shows that aircraft systems are highly integrated. Fuel systems, fire protection, avionics and electrical systems must operate reliably under vibration, temperature variation, altitude changes and emergency conditions. Proper redundancy, monitoring, maintenance and regulatory compliance are essential for safe aircraft operation.

References

[1] Federal Aviation Administration, *AC 20-42D: Hand Fire Extinguishers for Use in Aircraft*. Available: https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentid/888491

[2] Federal Aviation Administration, *AC 20-135: Powerplant Installation and Propulsion System Component Fire Protection Test Methods, Standards, and Criteria*. Available: https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/22194

[3] Federal Aviation Administration, *Aeronautical Information Manual: Performance-Based Navigation and Area Navigation*. Available: https://www.faa.gov/air_traffic/publications/atpubs/aim_html/chap1_section_2.html

[4] European Union Aviation Safety Agency, *CS-29 Large Rotorcraft Certification Specifications*. Available: https://www.easa.europa.eu/en/document-library/certification-specifications/group/cs-29-large-rotorcraft

[5] ASTM International, *ASTM D7566: Standard Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons*. Available: https://www.astm.org/Standards/D7566.htm

[6] Federal Aviation Administration, *AC 43.13-1B: Acceptable Methods, Techniques, and Practices – Aircraft Inspection and Repair, Chapter 11 Aircraft Electrical Systems*. Available: https://raanz.org.nz/techproject/FAA%20AC43.13%20b/Chapter%2011.pdf

[7] International Air Transport Association, *SAF Handbook*. Available: https://www.iata.org/en/programs/sustainability/reports/saf-handbook/section-2.1

[8] ExxonMobil, *Aviation Gasoline Product Data Sheet*. Available: https://www.exxonmobil.com/en-se/commercial-fuel/pds/gl-xx-avgas-series